Bill of Materials for an Electric Fixed-Wing RC Aircraft

Single-motor • Wingspan 1.0–1.2 m • Conventional / High-wing configuration
Functional electric propulsion • Wireless radio control • Lightweight repairable structure

Assumptions: *Target AUW 700–900 g • Hand-launch or short ground roll • 3S LiPo power system • Beginner-to-intermediate trainer • Prices approximate (INR, India market 2025–26). Specific example purchase links are listed in Part B and the Sources section.*

Part A — Component Identification

A complete functional electric fixed-wing aircraft requires the following major systems and materials:

Category	Components Included
Airframe & Structure	Foam sheets (Depron / EPP), carbon or bamboo spars, plywood / balsa reinforcement, control linkages, control horns, hinges
Propulsion	Brushless outrunner motor, Electronic Speed Controller (ESC) with BEC, propeller, propeller adapter / spinner
Power	LiPo battery pack, XT60 / bullet connectors, charge lead, battery strap / tray
Radio Control	2.4 GHz transmitter, matching receiver, binding plug (if required)
Actuation	Micro servos (aileron / elevator / rudder / throttle via ESC), servo extensions, pushrods, clevises
Landing Gear (optional)	Wire undercarriage or foam skid, wheels (if ground takeoff desired)
Wiring & Hardware	Motor wires, servo leads, heat-shrink, zip ties, screws, nuts, CA glue, epoxy, packing tape
Assembly Materials	Hot-wire / craft knife, sandpaper, rulers, soldering consumables, thread-lock

Control surfaces: Ailerons (roll), Elevator (pitch), Rudder (yaw). Throttle managed by ESC. Conventional tail or high-wing trainer layout selected for stability and ease of repair.

Part B — Comparison of Options (with Source Links)

For every major component at least two practical alternatives are listed with key specifications, quantity, approximate India-market price (INR) and example purchase links.

1. Motor

Component	Recommendation 1	Recommendation 2	Key Specs	Qty	Est. ■	Example Link
Motor	A2212 1000 KV outrunner (common clone)	D2830 / 2212 1000 KV (higher quality)	1000 KV, 2–3S, max ~12–15 A, prop 9×5–10×6, wt ~48–55 g	1	R1: 350–450 R2: 500–700	R1: robocraze.com/products/a2212-1000kv-bldc-brushless-motor R2: amazon.in (search D2830 1000KV)

2. ESC

Component	Recommendation 1	Recommendation 2	Key Specs	Qty	Est. ■	Example Link
ESC	30 A SimonK / generic with 5 V BEC	Hobbywing / DYS 30 A with SBEC	30 A cont., 40 A burst, 2–4S, BEC 5 V/2–3 A, wt ~20–25 g	1	R1: 350–500 R2: 700–1100	R1: amazon.in (30A ESC) R2: evelta.com (DYS-30A)

3. Battery

Component	Recommendation 1	Recommendation 2	Key Specs	Qty	Est. ■	Example Link
Battery	3S 2200 mAh 25C LiPo (generic / Orange)	Gens Ace / Turnigy 3S 2200 mAh 25–30C	11.1 V, 2200 mAh, 25C (55 A), XT60, wt ~165–180 g	1	R1: 1100–1400 R2: 1800–2200	R1: indiamart / local R2: rcmumbai.com (Gens Ace)

4. Propeller

Component	Recommendation 1	Recommendation 2	Key Specs	Qty	Est. ■	Example Link
Propeller	10×6 APC-style electric (Gemfan / clone)	Genuine APC 10×6E Thin Electric	10" dia × 6" pitch, electric series, balanced	2	R1: 80–150 ea R2: 250–350 ea	R1: xyzhobby.com / local R2: specialty RC shops

5. Servos

Component	Recommendation 1	Recommendation 2	Key Specs	Qty	Est. ■	Example Link
Servos	SG90 9 g plastic gear	MG90S 9 g metal gear	~1.8 kg·cm @ 4.8 V, 0.1 s/60°, 9–13 g	3–4	R1: 80–120 ea R2: 150–250 ea	R1: rees52.com / amazon R2: zbotic.in / robokits

6. Transmitter & Receiver

Component	Recommendation 1	Recommendation 2	Key Specs	Qty	Est. ■	Example Link
Tx + Rx	FlySky FS-i6 + iA6B (6 ch)	Radiomaster / FrSky entry 6–8 ch	2.4 GHz AFHDS 2A, 6 ch, range >500 m, Tx ~400 g	1 set	R1: 4200–5500 R2: 6000–9000	R1: robu.in / sunrobotics.in R2: specialty RC stores

7. Structural Materials

Component	Recommendation 1	Recommendation 2	Key Specs	Qty	Est. ■	Example Link
Foam	Depron / XPS 5 mm sheets	EPP 5–10 mm sheets / blocks	Light, easy cut; EPP more impact-resistant	4–6 sheets	R1: 800–1500 R2: 1500–3000	R1: stemvolt.in / vortex-rc.com R2: vortex-rc.com / indiamart
Spars	Carbon rod/tube 3–4 mm + bamboo	Balsa + plywood formers	Wing spar & fuselage longerons	As nee ded	300–600	Local hobby / online carbon suppliers

8. Other Essential Items

Item	Options / Notes	Qty	Est. ■	Example Source
Connectors (XT60, 3.5 mm bullets)	Pre-soldered preferred	Set	100–200	Robu.in / Amazon.in
Control linkages (pushrods, clevis, horns)	Metal or nylon	Set	150–300	Local RC / Amazon
Adhesives (CA, epoxy, foam-safe)	Hot glue also useful	As needed	200–400	Hardware / Amazon
Landing gear (optional wire + wheels)	Or simple foam skid	1 set	150–400	RC India forums / local
Heat-shrink, zip-ties, tape, screws	Workshop consumables	Assorted	150–300	Any electronics store

Primary Sources & Purchase Links (clickable in most PDF readers)

- Motor (A2212 1000 KV): https://robocraze.com/products/a2212-1000kv-bldc-brushless-motor
- ESC (DYS 30 A example): https://evelta.com/30a-2-4s-brushless-esc-with-5v-2a-bec-with-xt60-3-5mm-plug-for-rc-drones/
- Battery (Gens Ace 2200 mAh 3S): https://rcmumbai.com/products/gens-ace-2200mah-3s-25c-lipo-battery-xt60
- Radio (FlySky FS-i6): https://robu.in/product/flysky-fs-i6-2-4g-6ch-ppm-rc-transmitter-with-fs-ia6b-receiver/
- Servo (MG90S): https://zbotic.in/product/mg90s-9g-servo-metal-gear-180-degrees-superior-quality/
- Depron sheets: https://stemvolt.in/product/depron-sheet-40-inch-x-25-inch-5mm/ & https://www.vortex-rc.com/
- General: Amazon.in, Robu.in, SunRobotics.in, RC India forums (rcindia.org)

Note: Links and prices are examples current around late 2025 / 2026 and may change. Always verify stock, seller ratings and shipping before ordering.

Part C — Justification of Selections

1. Motor–Propeller Match and Thrust Check

Selected motor: A2212 1000 KV on 3S (11.1 V nominal). Static RPM ≈ 1000 × 11.1 ≈ 11 100 rpm (no-load). With a 10×6 prop the expected loaded current is typically 10–14 A and static thrust is approximately 700–900 g (manufacturer and community data for this common combination).

Target aircraft AUW is 700–900 g. Rule-of-thumb for a modest trainer: thrust-to-weight ratio ≥ 0.8–1.0 for comfortable hand-launch and climb. 700–900 g thrust on an 800 g airframe therefore meets the requirement with a small margin.

Practical check: Suspend the motor + prop on a digital scale (or use a simple thrust stand). Record current with a watt-meter and confirm that continuous current stays below motor and ESC ratings. Also verify propeller balance to minimise vibration.

2. ESC Rating and Electrical Margin

Measured or expected continuous current with the chosen prop is ~12 A. A 30 A ESC provides more than 2× margin. This covers short bursts, warm ambient temperatures, and gradual degradation of battery internal resistance. The integrated BEC (5 V / 2–3 A) is sufficient to power three or four 9 g servos (typical draw < 1 A total under normal flight loads).

3. Battery Selection and Trade-offs

3S 2200 mAh 25C is a widely available sweet-spot pack. Energy ≈ 24 Wh. At an average flight current of 8–10 A the theoretical duration is 13–16 minutes; practical flight time with reserve is 8–12 minutes.

Trade-offs: Higher capacity (e.g. 3000 mAh) increases flight time but adds mass and may require a larger wing or reduce climb performance. Higher C-rating (40C+) costs more and is unnecessary for this current draw. 4S would raise power but demands a lower-KV motor or smaller prop and increases structural loads.

4. Servo Count and Control Surfaces

Minimum functional set: three servos — aileron (or two aileron servos for differential), elevator, rudder. Throttle is handled by the ESC. A four-servo configuration (two aileron + elevator + rudder) is preferred for cleaner geometry and redundancy. 9 g metal-gear servos (MG90S) are recommended for durability; plastic-gear SG90s are acceptable for pure cost minimisation on a light airframe.

5. Structural Material Choice

Primary material: 5 mm Depron / XPS foam. Advantages — very low density, easy to cut and sand, inexpensive, adequate strength for 1 m class when reinforced with carbon or bamboo spars. EPP is superior for crash resistance and repairability (dents can often be heat-formed back) but is denser and more expensive. A hybrid approach (Depron wings + EPP fuselage nose) is also viable. Carbon spars or thin plywood formers provide the necessary bending stiffness while keeping mass low.

6. Electrical and Mechanical Compatibility

• Motor–ESC–Battery: All rated for 3S. ESC continuous current > expected load. Connectors matched (XT60 battery side, 3.5 mm bullets motor side).
• Transmitter–Receiver: FlySky FS-i6 ships bound to its iA6B receiver (AFHDS 2A protocol). Channels required: throttle, aileron, elevator, rudder (and optional dual aileron). Fail-safe must be programmed to cut throttle.
• Mechanical: Motor mount spacing (16/19 mm) matches common A2212 cross-mounts. Prop adapter fits 3.17 mm shaft. Servo arms and control horns selected for appropriate throw geometry; end-point and sub-trim adjusted on the transmitter.

Part D — Budget

1. Estimated Total Build Cost

Item Group	Budget Config (■)	Performance Config (■)
Motor	400	650
ESC 30 A	400	900
Battery 3S 2200 mAh	1200	2000
Propellers (2)	200	500
Servos (4)	400	800
Tx + Rx set	4500	7000
Foam + spars + hardware	1500	2800
Connectors, glue, linkages, misc.	600	900
Approximate Total	≈ **9 200**	≈ **15 550**

2. Safe Cost-Saving Opportunities

• Use plastic-gear SG90 servos instead of metal-gear (saves ~■400–600).
• Choose generic / clone motor and ESC from reputable local sellers with recent reviews.
• Build from Depron rather than full EPP; reinforce critically with carbon rod only where needed.

• Share or borrow a transmitter if one is already available in the club / lab (largest single cost).
• Buy propellers and small hardware in bulk or from forum group buys.
Avoid: extremely cheap unbranded LiPo batteries (safety risk) and motors with no published current data.

3. Budget-oriented vs Performance-oriented Configurations

Budget-oriented: A2212 1000 KV + generic 30 A ESC + 3S 2200 mAh 25C + SG90 servos + Depron airframe + FlySky FS-i6. Total ≈ ■9 000–10 000. Adequate for learning, short flights, and demonstration of all required functions.

Performance-oriented: Higher-quality 2212/2830 motor + branded ESC + Gens Ace battery + MG90S metal-gear servos + EPP or hybrid airframe + better radio. Total ≈ ■15 000–16 500. Better efficiency, durability, and longer service life.

4. Chosen Configuration for the Competition

The **budget-oriented configuration** is selected. Justification: It fully satisfies the functional requirements (electric single-motor, wireless control, 1–1.2 m wingspan, repairable structure, hand or ground launch) while keeping cost low and components easy to source and replace. For a student assignment the emphasis is on correct system matching and documentation rather than maximum performance. The chosen power system still provides a comfortable thrust-to-weight ratio.

Bonus — Take-off Mass and Propulsion Sufficiency

Estimated mass breakdown (budget configuration):

Item	Mass (g)
Airframe (foam + spars + covering)	280–350
Motor + mount + prop	70–80
ESC	25
Battery 3S 2200 mAh	170–180
Servos (4 × 10 g) + linkages	55–65
Receiver + wiring + hardware	40–50
Landing gear / skid (optional)	20–40
Total AUW (typical)	**660–790 g**

Expected static thrust with A2212 1000 KV + 10×6 prop on 3S: approximately 750–900 g. Thrust-to-weight ratio at 750 g AUW ≈ 1.0–1.2. This is sufficient for comfortable hand-launch and a positive climb rate. At the upper mass limit (~800 g) the ratio remains ≥ 0.9, still acceptable for a trainer-style aircraft.

Assumption summary: Sea-level density, propeller in good condition, battery at full charge, motor run within continuous rating. Actual thrust should be verified on a simple stand before first flight.

Summary BOM (Selected Budget Configuration)

#	Component	Specification	Qty	Est. ■	Example Source
1	Brushless motor	A2212 1000 KV	1	400	robocraze.com
2	ESC	30 A with 5 V BEC	1	400	Amazon / Evelta
3	Battery	3S 2200 mAh 25C LiPo	1	1200	rcmumbai.com / local
4	Propeller	10×6 electric (2 pcs)	2	200	Local / xyzhobby
5	Servos	SG90 9 g	4	400	rees52 / Amazon
6	Radio set	FlySky FS-i6 + iA6B	1	4500	robu.in
7	Airframe materials	Depron + carbon/bamboo + hardware	—	1500	vortex-rc / stemvolt
8	Misc. (glue, connectors, linkages)	Assorted	—	600	Various
	TOTAL			≈ **9 200**	

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All prices and links are approximate market examples for educational purposes only. Always cross-check current availability, safety ratings and local regulations before purchasing or flying.